

Microsoft Entra ID IAM & Identity Governance Lab

Project Type: Identity and Access Management Lab

Platform: Microsoft Entra ID

Organization: Microserve

Application: Microserve Expense Portal

1. Project Overview

1.1 Project Objective

The objective of this project is to design and implement a practical Identity and Access Management (IAM) environment using Microsoft Entra ID.

The project demonstrates how an organization can manage identities, control access to applications, govern user access, protect privileged accounts, enforce authentication security, and monitor identity-related activities.

The following Microsoft Entra capabilities are demonstrated:

- User and group management
- Role-Based Access Control (RBAC)
- Application registration and application roles
- Conditional Access
- Multi-Factor Authentication (MFA)
- Identity Governance
- Entitlement Management
- Access Packages
- Access Reviews
- Separation of Duties
- Privileged Identity Management (PIM)
- Device identity and device settings
- Identity Protection
- Sign-in and audit monitoring

1.2 Business Scenario

Microserve is a fictional organisation that uses an internal **Expense Portal** application.

The Expense Portal requires different levels of access for employees and users responsible for approving expenses.

The project implements two application roles:

- **Expense User**
- **Expense Approver**

Access to the application is governed using Microsoft Entra ID. Users can request access through an Access Package, requests can require approval, access can expire, and access can subsequently be reviewed.

Privileged administrative access is separately controlled using Microsoft Entra Privileged Identity Management (PIM).

1.3 Project Goals

The main goals of the project are to:

1. Implement application-based authorization using Entra application roles.
2. Control access through groups and role assignments.
3. Implement an Access Package for governed application access.
4. Configure an approval-based access request workflow.
5. Implement Access Reviews for ongoing access governance.
6. Configure privileged access using PIM.
7. Require MFA and justification for privileged role activation.
8. Configure Conditional Access policies.
9. Review device identity and device-management settings.
10. Use sign-in logs, audit logs, PIM history, and Identity Protection for monitoring and investigation.

2. Lab Environment

2.1 Microsoft Entra ID Tenant

The project was implemented in a dedicated Microsoft Entra ID lab environment representing the fictional organization **Microserve**.

The lab environment was used to configure and test identity, access management, identity governance, privileged access, Conditional Access, device settings, and security monitoring capabilities.

2.2 Application

The primary application used in the lab was:

Microserve Expense Portal

The application was used as the business scenario for demonstrating application authorization and governed access.

Two application roles were configured:

Application Role	Purpose
Expense User	Provides standard access to the Expense Portal
Expense Approver	Provides access to expense approval functionality

2.3 Test Users

The following identities were used during testing:

User	Purpose
Alex Martin	Used to test Expense Portal access, Access Package requests, and user access workflows
John Smith	Used to test privileged access through Privileged Identity Management
Satya Rellu	Used as the administrative account for configuring the lab environment

2.4 Groups

The following group was created:

Group	Purpose
Expense.user	Contains users who are permitted to request access to the Expense Portal Access Package

Alex Martin was added as a member of the **Expense.user** group for testing the access request workflow.

2.5 Major Microsoft Entra Components

The following Microsoft Entra capabilities were used in the project:

- **App registrations**
- **Enterprise applications**
- **Application roles**
- **Groups**
- **Role-Based Access Control (RBAC)**
- **Conditional Access**
- **Identity Governance**
- **Entitlement Management**
- **Access Packages**
- **Access Reviews**
- **Privileged Identity Management (PIM)**
- **Device settings**
- **Identity Protection**
- **Sign-in Logs**
- **Audit Logs**

Microserve Expense Portal application registration in Microsoft Entra ID

The screenshot displays the Microsoft Entra admin center interface. The left sidebar contains navigation options such as Users, Groups, Devices, Agents, Enterprise apps, App registrations, Roles & admins, Delegated admin partners, Tenant governance, Domain services, Conditional Access, Multifactor authentication, Identity Secure Score, Authentication methods, Account recovery, Password reset, Custom security attributes, Certificate authorities, External Identities, Cross-tenant synchronization, Entra Connect, and Backup and recovery. The main content area shows the 'Microserve Expense Portal' application registration details. The 'Essentials' section includes fields for Display name, Application (client) ID, Object ID, Directory (tenant) ID, Supported account types, Client credentials, Redirect URIs, Application ID URI, Managed application in L, and State. The 'Get Started' section provides links to 'Build your application with the Microsoft identity platform', 'Call APIs', 'Sign in users in 5 minutes', and 'Configure for your organization'.

Microsoft Entra admin center

Search resources, services, and docs (G+)

Users · Usage & insights | Microsoft Entra application activity (Preview) · Bulk Operations · Audit Logs · Provisioning Logs · Security | Conditional Access · Conditional Access | Insights and reporting · Sign-in events · App registrations

Microserve Expense Portal

Deactivate Delete Endpoints

Get a second? We would love your feedback on Microsoft identity platform (previously Azure AD for developer). →

Essentials

Display name	: Microserve Expense Portal	Client credentials	: Add a certificate or secret
Application (client) ID	: XXXXXXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXXXX	Redirect URIs	: Add a Redirect URI
Object ID	: XXXXXXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXXXX	Application ID URI	: Add an Application ID URI
Directory (tenant) ID	: XXXXXXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXXXX	Managed application in L	: Microserve Expense Portal
Supported account types	: My organization only	State	: Activated

Get Started Documentation

Build your application with the Microsoft identity platform

The Microsoft identity platform is an authentication service, open-source libraries, and application management tools. You can create modern, standards-based authentication solutions, access and protect APIs, and add sign-in for your users and customers. [Learn more](#)

Call APIs
Build more powerful apps with rich user and business data from Microsoft services and your own company's data sources.
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Use our SDKs to sign in users and call APIs in a few steps. Use the quickstarts to start a web app, mobile app, SPA, or daemon app.
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Configure for your organization
Assign users and groups, apply conditional access policies, configure single sign-on, and more in Enterprise applications.
[Go to Enterprise applications](#)

3. Identity and Access Management

3.1 User Identity Management

Microsoft Entra ID was used to create and manage test identities for the Microserve environment. Multiple test accounts were created to represent different users and administrative scenarios.

The primary test users used throughout the project were:

- **Alex Martin:** Used to test application access, group membership, access package requests, and access governance.
- **John Smith:** Used to test privileged access and Privileged Identity Management (PIM).
- **Satya Rellu:** Used as the administrative/configuration account for implementing and validating the lab.

These accounts provided separate identities for testing standard user access, privileged access, and administrative activities.

The screenshot displays the Microsoft Entra admin center interface. The left sidebar shows the navigation menu with options like Home, Favorites, Entra ID, Overview, Users, Groups, Devices, Agents, Enterprise apps, App registrations, Roles & admins, Delegated admin partners, Tenant governance, Domain services, Conditional Access, Multifactor authentication, Identity Secure Score, Authentication methods, Account recovery, and Password reset. The main content area is titled 'Users' and shows a list of 14 users. The table includes columns for Display name, User principal name, User type, Is Agent, On-premises sync, Sign-in identifiers, and Company name. The users listed are Alex Martin, Bindu Sai, Daniel Clark, David Brown, Emma Taylor, Greg Johnson, John, John Smith, Lisa Chen, Mark Wilson, Nike, Robert, Sarah Wilson, and Satya Rellu.

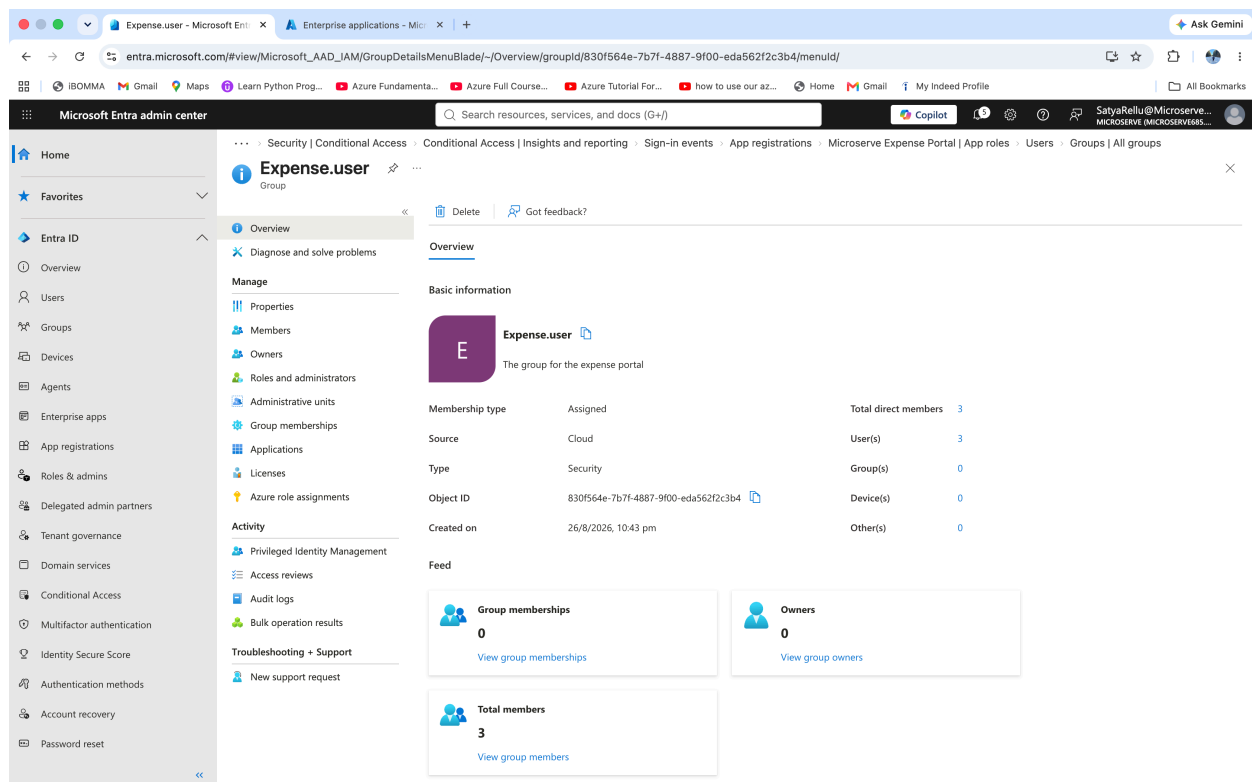
	Display name ↑	User principal name ↑	User type	Is Agent	On-premises sync	Sign-in identifiers	Company name
☐	AM Alex Martin	AlexMartin@Micro...	Member	No	No	View	Microserve
☐	B Bindu Sai	BinduSai@Micro...	Member	No	No	View	
☐	DC Daniel Clark	DanielClark@Micro...	Member	No	No	View	Microserve
☐	DB David Brown	DavidBrown@Micro...	Member	No	No	View	Microserve
☐	ET Emma Taylor	EmmaTaylor@Micro...	Member	No	No	View	
☐	GJ Greg Johnson	GregJohnson@Micro...	Member	No	No	View	
☐	J John	John@Microserve...	Member	No	No	View	
☐	JS John Smith	Johnsmith@Micro...	Member	No	No	View	Microserve
☐	LC Lisa Chen	Lisachen@Micro...	Member	No	No	View	Microserve
☐	MW Mark Wilson	MarkWilson@Micro...	Member	No	No	View	Microserve
☐	N Nike	Nike@Microserve...	Member	No	No	View	
☐	R Robert	robert@Microserv...	Member	No	No	View	
☐	SW Sarah Wilson	SarahWilson@Micro...	Member	No	No	View	Microserve
☐	SR Satya Rellu	SatyaRellu@Micro...	Member	No	No	View	

3.2 Group-Based Access

A security group named **Expense.user** was created to manage access to the Microserve Expense Portal.

Alex Martin was added as a member of the group. The group was subsequently used in access-control scenarios, including application access and Conditional Access configuration.

Using a group rather than assigning controls individually to each user demonstrates a scalable approach to identity and access management.



The screenshot displays the Microsoft Entra admin center interface. The left-hand navigation pane includes sections for Home, Favorites, and a list of administrative tools such as Entra ID, Users, Groups, Devices, Agents, Enterprise apps, App registrations, Roles & admins, Delegated admin partners, Tenant governance, Domain services, Conditional Access, Multifactor authentication, Identity Secure Score, Authentication methods, Account recovery, and Password reset. The main content area is titled 'Expense.user' and shows the 'Overview' tab. Under the 'Manage' section, there are links for Properties, Members, Owners, Roles and administrators, Administrative units, Group memberships, Applications, Licenses, and Azure role assignments. The 'Activity' section includes links for Privileged Identity Management, Access reviews, Audit logs, Bulk operation results, and Troubleshooting + Support. The 'Basic information' section provides details about the group: Membership type is Assigned, Source is Cloud, Type is Security, Object ID is 830f564e-7b7f-4887-9f00-eda562f2c3b4, and it was created on 26/8/2026 at 10:43 pm. A table on the right shows membership counts: Total direct members (3), User(s) (3), Group(s) (0), Device(s) (0), and Other(s) (0). At the bottom, there are three summary cards: 'Group memberships' with 0 items, 'Owners' with 0 items, and 'Total members' with 3 items. Each card has a 'View group...' link.

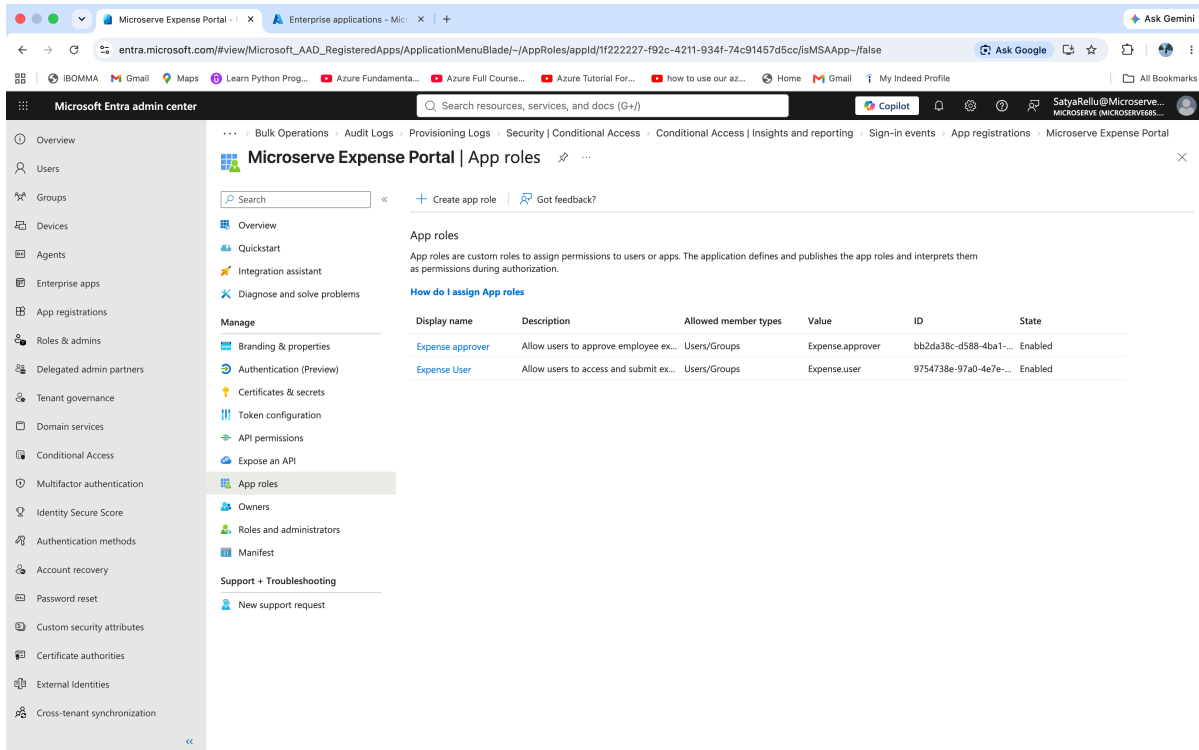
3.3 Role-Based Access Control

Role-based access was implemented at both the application and Microsoft Entra administrative levels. The **Microserve Expense Portal** was configured with two application roles:

- **Expense User**
- **Expense Approver**

These roles represent separation of responsibilities within the application. An Expense User represents a standard employee who requires access to expense-related functionality, while an Expense Approver represents a user responsible for reviewing or approving expense submissions.

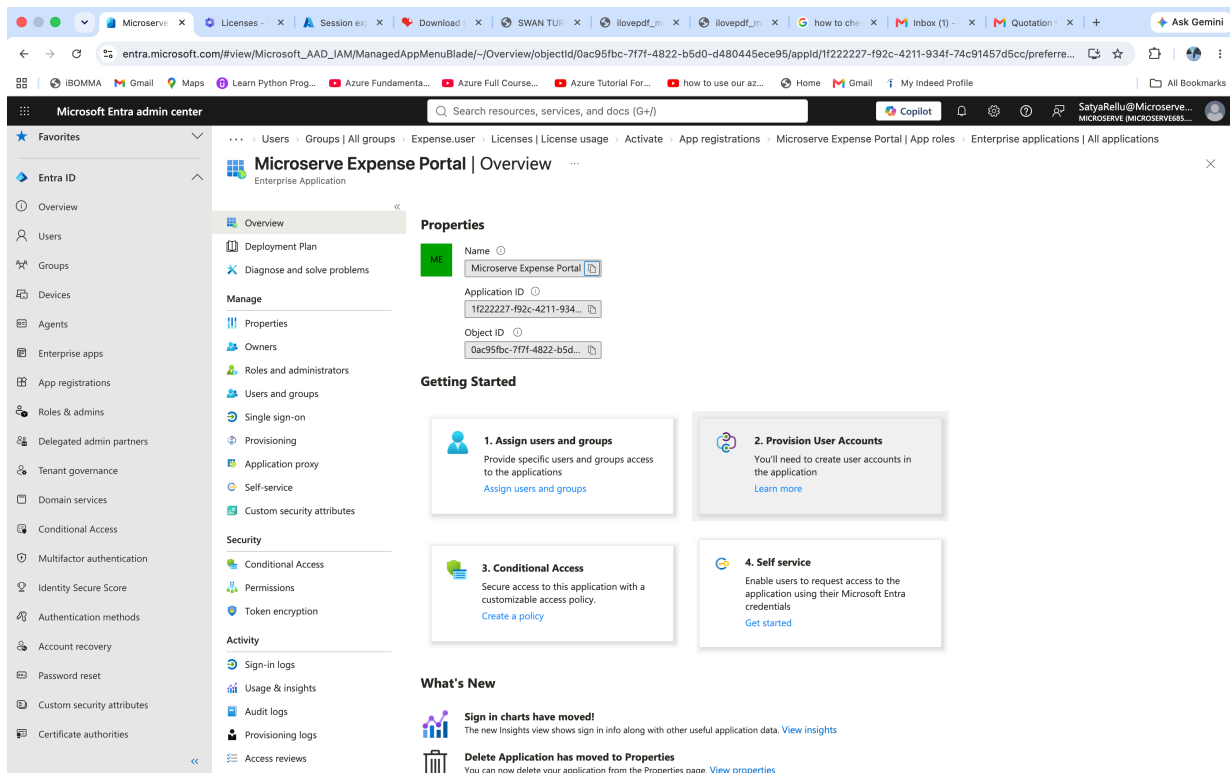
Microsoft Entra administrative roles were also used during the project to demonstrate privileged access management. The **Security Administrator** role was selected for PIM testing.



4. Application Registration & Enterprise Applications

This section documents how the **Microserve Expense Portal** was registered in Microsoft Entra ID and made available as an enterprise application.

4.1 Application Registration



The Microserve Expense Portal was registered as an application in Microsoft Entra ID to establish an identity for the application and enable Microsoft Entra-based authentication and authorization.

The application registration provides the application with a unique client ID and establishes its relationship with the Microserve tenant.

The application was configured as a single-tenant application, allowing it to be used within the organization's Microsoft Entra environment.

4.2 Application Roles

Two application roles were configured for the Microserve Expense Portal:

- **Expense User**
- **Expense Approver**

Application roles provide a mechanism for assigning users different levels of application access based on their responsibilities.

The Expense User role represents standard access to the expense application, while the Expense Approver role represents users responsible for approving expense-related requests.

4.3 Enterprise Application

The application registration resulted in an associated enterprise application within Microsoft Entra ID. The enterprise application provides the administrative interface for managing access to the application, including users, groups, assignments, and authentication-related settings.

This demonstrates the distinction between **App registrations**, which define the application identity and configuration, and **Enterprise applications**, which provide tenant-level access and management for the application.

4.4 Application Access Model

Access to the Microserve Expense Portal was designed around group membership and application roles. The **Expense.user** security group was used as the primary user-access group, while application roles provided role-specific authorization.

This approach provides a scalable access model in which users can be managed through groups and assigned application permissions according to their responsibilities.

5. Conditional Access

5.1 Conditional Access Overview

Microsoft Entra Conditional Access was used to implement policy-based access controls for the Microserve environment. Conditional Access evaluates signals such as the user, application, device, location, client application, and risk before determining whether access should be allowed or restricted.

The Conditional Access configuration in this lab demonstrates how organizations can apply Zero Trust principles by requiring additional controls when specific access conditions are met.

5.2 Conditional Access Policies

Multiple Conditional Access scenarios were configured and evaluated during the project. The policies were designed to demonstrate controls such as multifactor authentication, application-specific access requirements, and device-based access conditions.

Policies were tested using the Microsoft Entra sign-in logs and the Conditional Access **What If** tool where applicable. Report-only mode was used where appropriate to evaluate the expected impact of a policy without immediately enforcing the control.

5.3 Require Compliant Device for Expense Portal

A Conditional Access policy named **CA010 - Require Compliant Device for Expense Portal** was created to demonstrate device-based access control.

The policy targets the **Expense.user** group and the **Microserve Expense Portal** application. The grant control requires the accessing device to be marked as compliant.

The policy was configured in **Report-only** mode to evaluate its potential effect without blocking users in the lab environment.

During validation, the policy could not be fully tested because the lab environment did not contain an Intune-managed compliant device. The Microsoft Entra tenant contained no registered devices at the time of testing. The policy configuration itself was successfully created and reviewed.

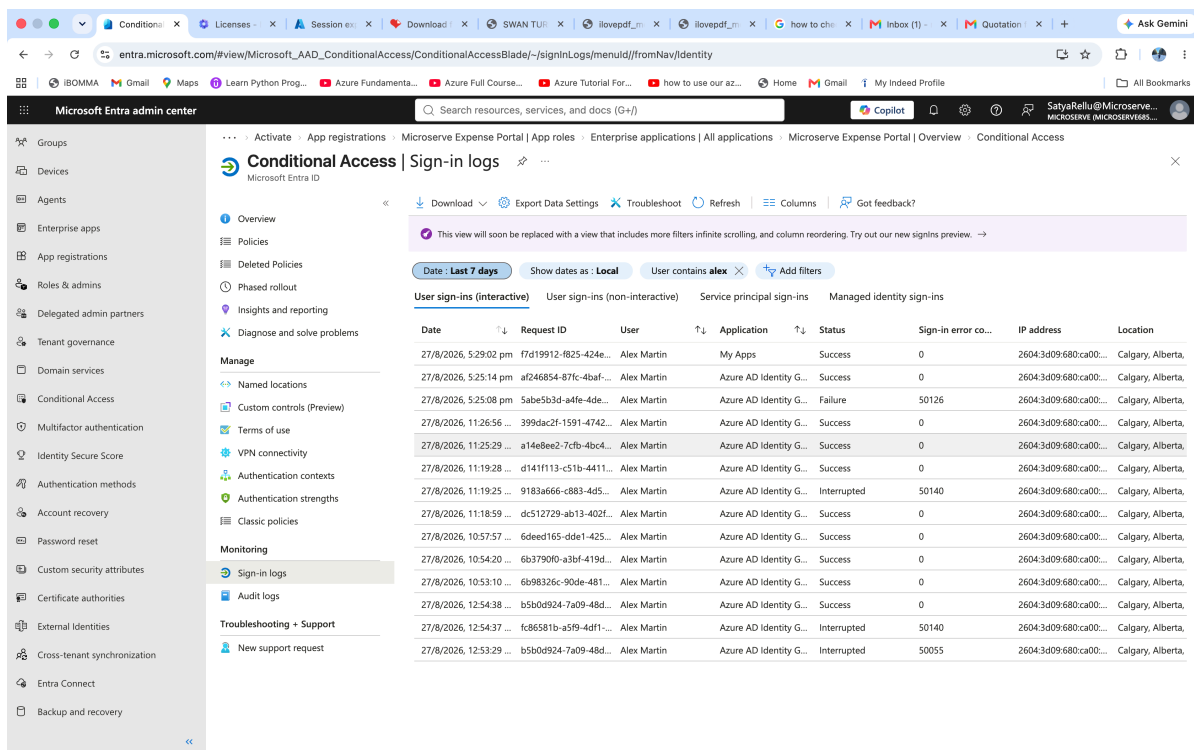
The screenshot displays the Microsoft Entra admin center interface. The left sidebar shows the navigation menu with categories like Groups, Devices, Agents, Enterprise apps, App registrations, Roles & admins, Delegated admin partners, Tenant governance, Domain services, Conditional Access, Multifactor authentication, Identity Secure Score, Authentication methods, Account recovery, Password reset, Custom security attributes, Certificate authorities, External Identities, Cross-tenant synchronization, Entra Connect, and Backup and recovery. The main content area is titled 'Conditional Access | Policies' and shows a list of 15 policies. The policy 'CA010 - Require Compliant Device for Expense Portal' is highlighted. The right pane shows the 'Policy details' for this policy, including its state (Report-only), users and workload identities (1 user, 1 group), included resources (1 resource), requirements for access (Require device to be marked as compliant), client apps (1 included), and modified date (27/08/26, 6:24:21 pm).

Policy name	Policy details
Block High Risk Android Users	
CA001-Require MFA for Administrators	
CA002- Require MFA for Expense portal	
CA003 configuration	
CA004-Require MFA for risky sign-ins	
CA005: Require Compliant Device	
CA006 - Require Phishing Resistant Authentication	
CA007 - Block access from untrusted locations	
CA008- Require approved client apps	
CA009-Device Platform Restrictions	
CA010 - Require Compliant Device for Expense Portal	
CA010-Require Risk Remediation for High-Risk Sign-ins	
Microsoft authentication Method	
Require MFA for User & Sign in	
block greg from access	

5.4 Conditional Access Validation

Conditional Access behavior was reviewed using Microsoft Entra sign-in logs and the What If analysis tool. Sign-in activity was examined to verify authentication requirements and Conditional Access evaluation where applicable.

Because the lab used a limited test environment and did not contain an Intune-managed compliant device, device-compliance enforcement could not be demonstrated through a successful end-to-end blocking scenario.



Date	Request ID	User	Application	Status	Sign-in error co...	IP address	Location
27/8/2026, 5:29:02 pm	f7d19912-f825-424e...	Alex Martin	My Apps	Success	0	2604:3d09:680:ca00...	Calgary, Alberta,
27/8/2026, 5:25:14 pm	a246854-87fc-4baf...	Alex Martin	Azure AD Identity G...	Success	0	2604:3d09:680:ca00...	Calgary, Alberta,
27/8/2026, 5:25:08 pm	5ab5b3d-a4fe-4de...	Alex Martin	Azure AD Identity G...	Failure	50126	2604:3d09:680:ca00...	Calgary, Alberta,
27/8/2026, 11:26:56 ...	399dac2f-1591-4742...	Alex Martin	Azure AD Identity G...	Success	0	2604:3d09:680:ca00...	Calgary, Alberta,
27/8/2026, 11:25:29 ...	a14e8ee2-7cfb-4bc4...	Alex Martin	Azure AD Identity G...	Success	0	2604:3d09:680:ca00...	Calgary, Alberta,
27/8/2026, 11:19:28 ...	d141f113-c51b-4411...	Alex Martin	Azure AD Identity G...	Success	0	2604:3d09:680:ca00...	Calgary, Alberta,
27/8/2026, 11:19:25 ...	9183a666-c883-4d5...	Alex Martin	Azure AD Identity G...	Interrupted	50140	2604:3d09:680:ca00...	Calgary, Alberta,
27/8/2026, 11:18:59 ...	dc512729-ab13-402f...	Alex Martin	Azure AD Identity G...	Success	0	2604:3d09:680:ca00...	Calgary, Alberta,
27/8/2026, 10:57:57 ...	6deed165-dde1-425...	Alex Martin	Azure AD Identity G...	Success	0	2604:3d09:680:ca00...	Calgary, Alberta,
27/8/2026, 10:54:20 ...	6b3790f0-a3bf-419d...	Alex Martin	Azure AD Identity G...	Success	0	2604:3d09:680:ca00...	Calgary, Alberta,
27/8/2026, 10:53:10 ...	6b98326c-90de-481...	Alex Martin	Azure AD Identity G...	Success	0	2604:3d09:680:ca00...	Calgary, Alberta,
27/8/2026, 12:54:38 ...	b5b0d924-7a09-48d...	Alex Martin	Azure AD Identity G...	Success	0	2604:3d09:680:ca00...	Calgary, Alberta,
27/8/2026, 12:54:37 ...	fc86581b-a5f9-4df1...	Alex Martin	Azure AD Identity G...	Interrupted	50140	2604:3d09:680:ca00...	Calgary, Alberta,
27/8/2026, 12:53:29 ...	b5b0d924-7a09-48d...	Alex Martin	Azure AD Identity G...	Interrupted	50055	2604:3d09:680:ca00...	Calgary, Alberta,

6. Authentication, MFA & Self-Service Password Reset

6.1 Multifactor Authentication

Multifactor authentication (MFA) was incorporated into the Microserve identity and access design to provide an additional layer of protection beyond username and password authentication.

MFA was used as part of the authentication and access-control scenarios implemented in the lab. Conditional Access policies can require MFA when users access protected resources or when specific conditions are met.

MFA was also incorporated into privileged access scenarios through Microsoft Entra Privileged Identity Management (PIM), providing additional protection when privileged roles are activated.

6.2 Authentication Methods

Microsoft Entra authentication methods were reviewed as part of the identity security configuration. Authentication methods provide users with mechanisms for proving their identity and support stronger authentication requirements such as multifactor authentication.

The lab also demonstrated how authentication requirements can be incorporated into Conditional Access and privileged-access controls rather than relying solely on a password.

6.3 Self-Service Password Reset

Self-Service Password Reset (SSPR) was reviewed as an identity-management capability that allows users to reset or change their passwords without requiring intervention from an administrator.

SSPR supports operational efficiency while reducing dependency on help-desk password-reset procedures. It can also be combined with authentication methods to verify a user's identity before allowing a password reset.

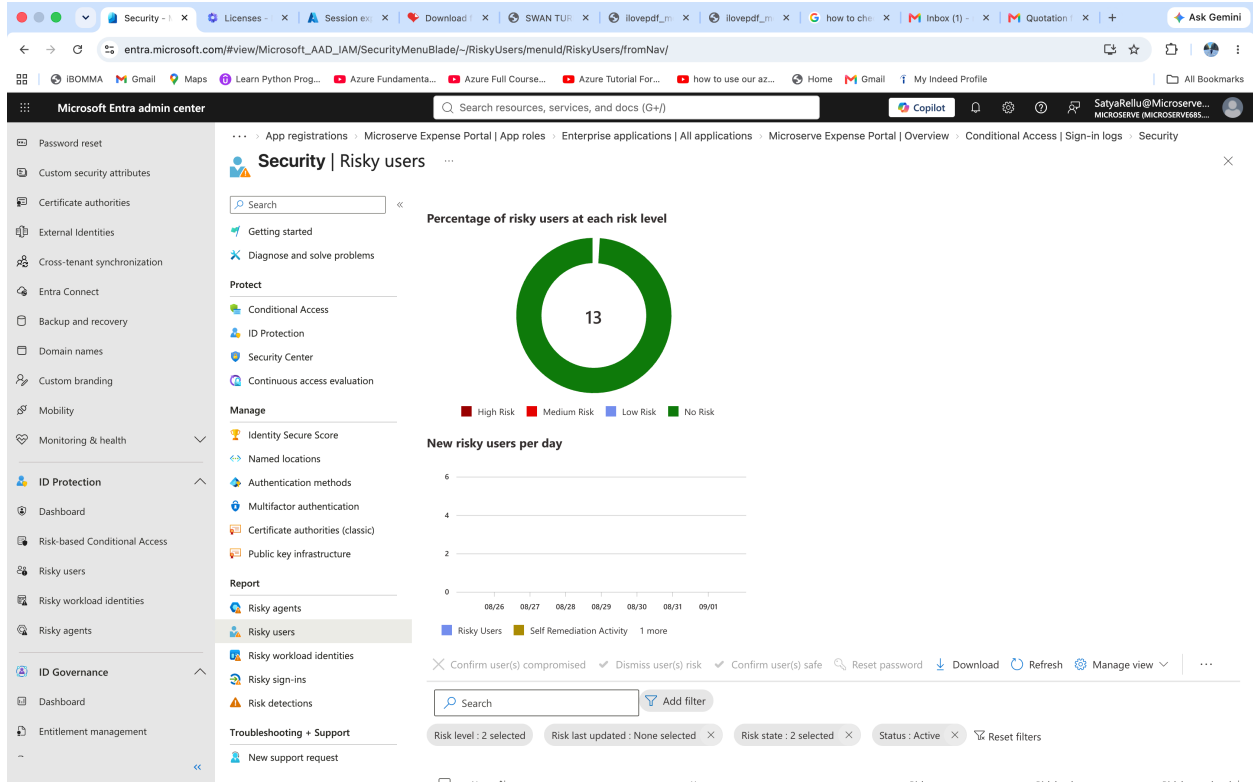
6.4 Authentication and Access Design

The authentication design in the Microserve environment follows a layered security approach. User authentication provides the initial identity verification, while MFA, Conditional Access, and privileged access controls provide additional security controls based on the context of the access request.

This approach supports the Zero Trust principle of continuously evaluating access rather than assuming that an authenticated user should automatically receive unrestricted access.

7. Identity Protection & Risk Management

7.1 Microsoft Entra Identity Protection



Microsoft Entra ID Protection was reviewed and used to demonstrate identity-risk monitoring within the Microserve environment. Identity Protection helps identify potentially compromised identities by analyzing risk signals associated with users and sign-ins.

The lab included reviewing the available **Risky users** and **Risky sign-ins** areas to understand how identity-related risks can be investigated and managed.

7.2 Risky Users

The Risky users experience was reviewed to understand how Microsoft Entra identifies users whose accounts may be compromised.

During testing, **no risky users were present in the lab environment**. This was expected because the lab did not intentionally generate simulated or detected account-compromise activity.

In a production environment, administrators could use this information to investigate affected accounts and apply appropriate remediation actions.

7.3 Risky Sign-ins

Risky sign-ins were also reviewed as part of the identity-protection investigation process.

No risky sign-ins were identified during the lab testing period. Normal successful sign-in activity was observed in the tenant, but the lab did not generate a sign-in event that Microsoft Entra classified as risky.

This allowed the investigation interface and risk-monitoring capabilities to be reviewed without introducing artificial security events into the environment.

7.4 Risk-Based Access Strategy

Identity Protection complements Conditional Access by providing risk signals that can be used when making access decisions.

A production implementation could use user risk and sign-in risk as Conditional Access conditions. For example, higher-risk authentication events could require multifactor authentication or additional remediation before access is permitted.

This demonstrates a risk-based approach to identity security in which access decisions are influenced not only by the user's identity but also by the security context surrounding the authentication attempt.

8. Identity Governance & Entitlement Management

8.1 Identity Governance Overview

Microsoft Entra Identity Governance was used to implement controlled, reviewable, and lifecycle-based access to resources within the Microserve environment.

The lab demonstrated how access can be requested, approved, assigned, reviewed, and eventually removed through Microsoft Entra Entitlement Management.

This approach helps organizations follow the principle of least privilege while providing a structured process for managing user access.

8.2 Entitlement Management

Microsoft Entra Entitlement Management was used to create a controlled access process for the Microserve Expense Portal.

An **Access Catalog** named **Expense Portal Catalog** was created to organize resources associated with the application.

An **Access Package** named **Expense report access** was then created within the catalog. The package included access to the Microserve Expense Portal and its configured application roles.

The access package provides a centralized mechanism for users to request access and for administrators to control and review those requests.

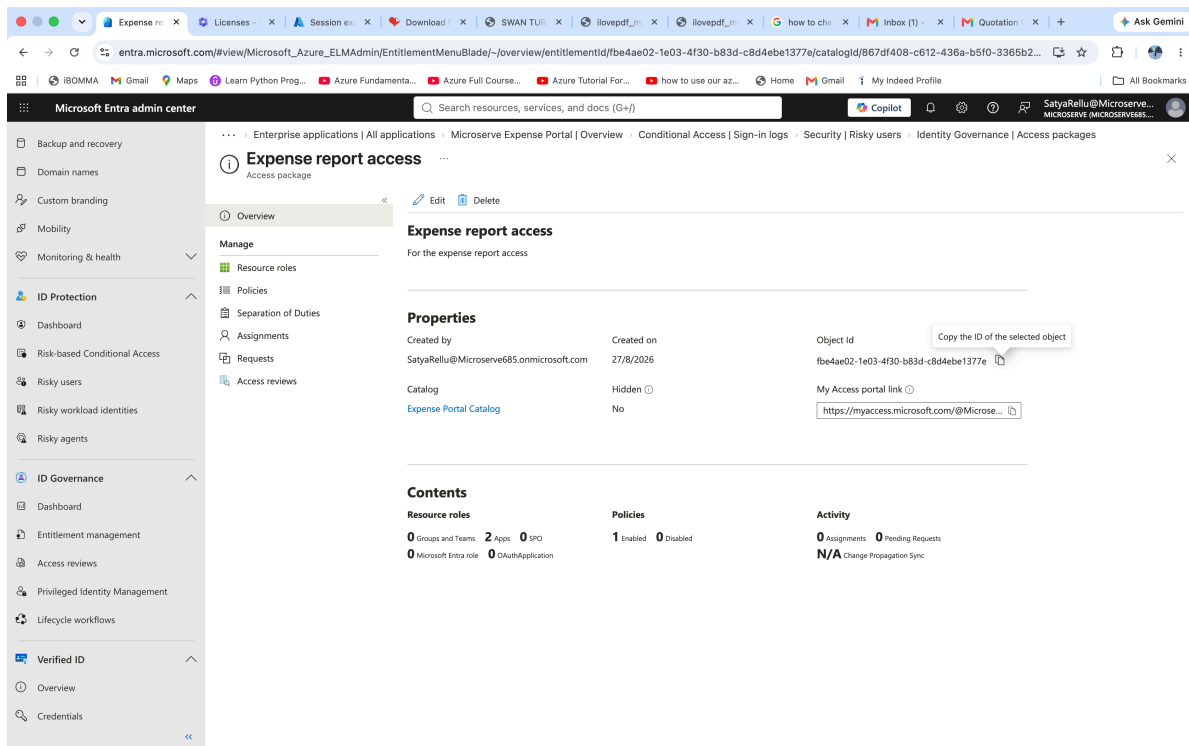
8.3 Access Package Policy

An access package policy named **Initial Policy** was configured to control how users could request access.

The policy was configured to allow users in the **Expense.user** group to request access. Self-service requests were enabled, and approval was required before access could be granted.

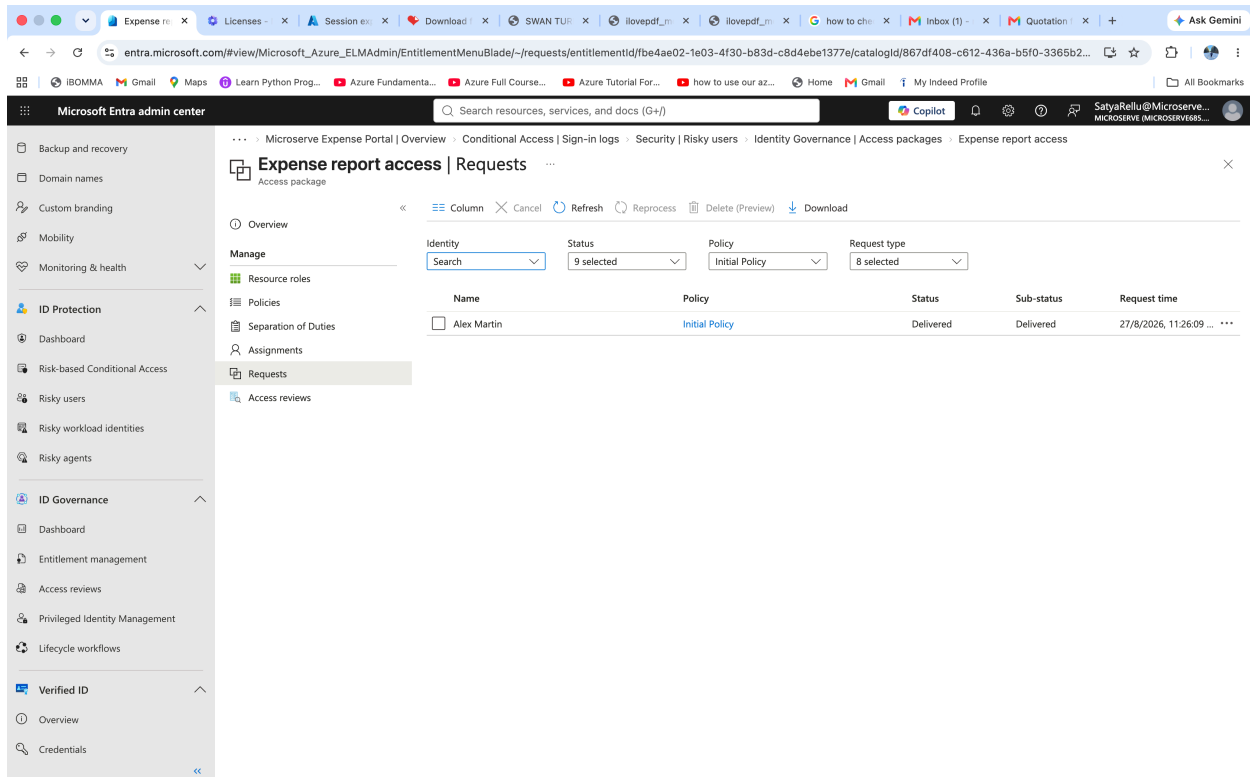
Requestors were also required to provide justification for their request. A custom question, "**why do you need access**", was configured to collect the business reason for requesting access.

The access assignment was configured with an expiration date of **September 29, 2026**, providing a defined access lifecycle rather than indefinite access.



8.4 Access Request and Approval Workflow

The access request workflow was tested using Alex Martin as the requestor.



Alex Martin submitted a request for the **Expense report access** access package through the My Access portal. The request required approval before the access could be assigned.

The request was subsequently reviewed and approved by the administrator. Following approval, Alex Martin appeared under the access package assignments, confirming that the governance workflow had successfully granted the requested access.

8.5 Access Reviews

Access Reviews were configured and reviewed as part of the identity governance implementation.

Access Reviews provide a mechanism for periodically validating whether users should continue to have access to organizational resources. This supports least-privilege access and helps prevent unnecessary access from remaining active over time.

The lab demonstrated the Access Review functionality associated with the access package.

8.6 Separation of Duties

Separation of Duties (SoD) was considered as part of the Expense Portal access design.

The application contains separate **Expense User** and **Expense Approver** roles. From a business perspective, these roles can represent different responsibilities, helping prevent a single individual from both submitting and approving their own expense transactions.

The lab used this role separation to demonstrate the concept of conflicting responsibilities. A full production implementation would require explicit SoD policies and business rules appropriate to the organization's approval process.

8.7 Governance Outcome

The Identity Governance implementation demonstrated an end-to-end controlled access lifecycle:

User → Access Request → Approval → Assignment → Access Review

The workflow provided a practical example of how Microsoft Entra can move application access away from informal, permanent assignments toward a controlled and auditable access-management process.

9. Privileged Identity Management (PIM)

9.1 PIM Overview

Microsoft Entra Privileged Identity Management (PIM) was implemented to provide controlled, time-bound access to privileged Microsoft Entra roles.

PIM reduces the need for users to maintain permanent privileged access by allowing privileged roles to be assigned as **eligible** and activated only when required.

The lab used PIM to demonstrate just-in-time privileged access, activation requirements, approval workflows, and auditability.

9.2 Eligible Role Assignment

John Smith was assigned as an eligible member of the **Security Administrator** role using Microsoft Entra PIM.

The assignment was configured as a time-bound eligible assignment rather than a permanent active role assignment. This allows the user to activate the privileged role when required while keeping the role inactive during normal operation.

Microsoft Entra admin center

Search resources, services, and docs (G+)

My roles | Microsoft Entra roles

Privileged Identity Management | My roles

Activate

Microsoft Entra roles

Groups

Azure resources

Troubleshooting + Support

Troubleshoot

New support request

Eligible assignments

Active assignments

Expired assignments

Role	Scope	Membership	State	End time	Action
Attribute Assignment Admin...	Microserve	Direct	Assigned	Permanent	Deactivate
Global Administrator	Microserve	Direct	Assigned	Permanent	Deactivate
Attribute Definition Adminis...	Microserve	Direct	Assigned	Permanent	Deactivate

This configuration demonstrates the principle of least privilege by limiting standing administrative access.

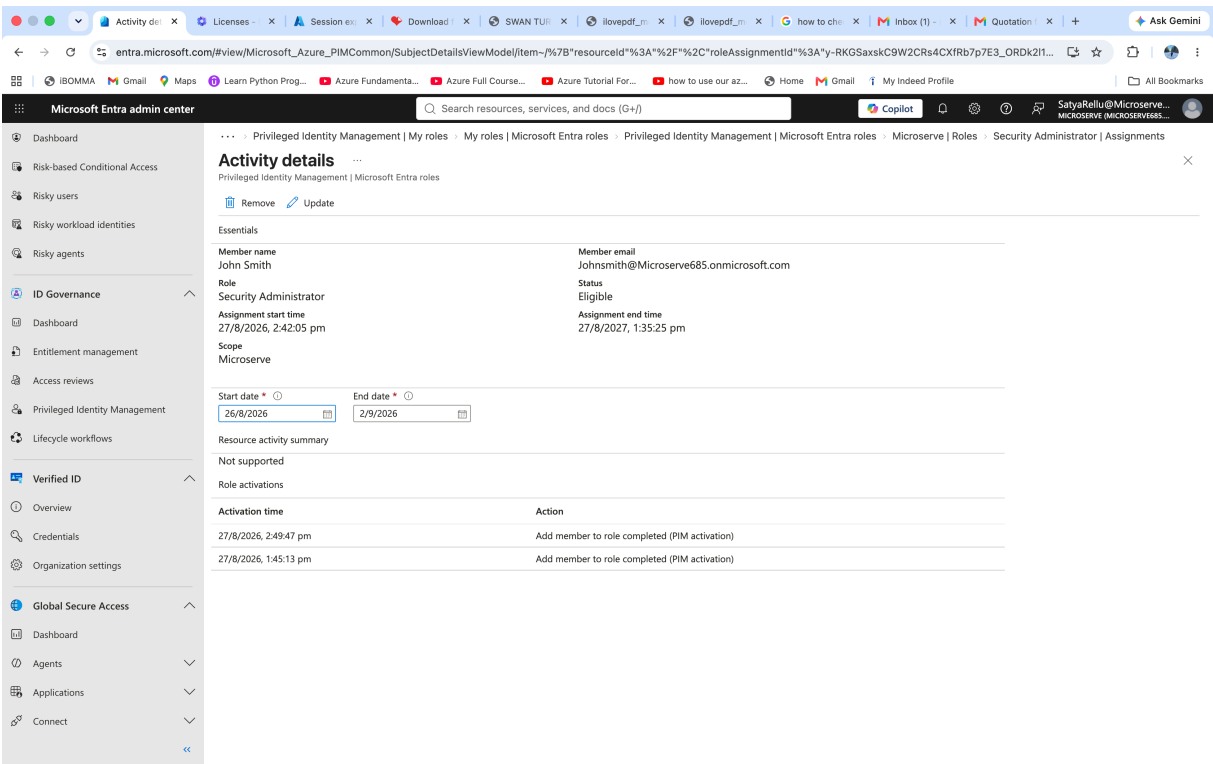
9.3 PIM Activation Controls

PIM activation controls were configured to require additional security measures before the eligible role could become active.

The activation configuration included:

- Maximum activation duration of **2 hours**
- **Multifactor authentication** required for activation
- **Justification** required when activating the role
- Approval required for privileged activation

These controls ensure that privileged access is temporary and that activation is subject to additional verification and accountability.

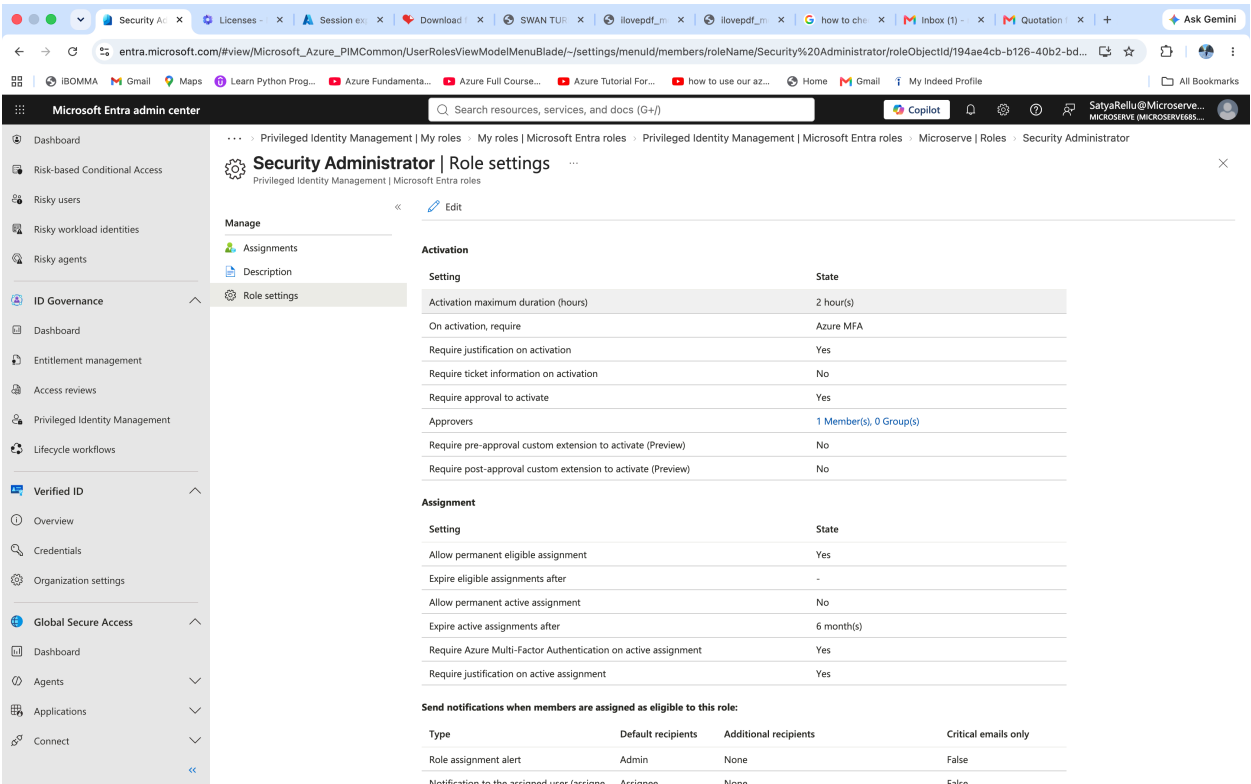


9.4 Privileged Role Activation

The PIM activation workflow was tested using John Smith.

John Smith requested activation of the eligible **Security Administrator** role. Because approval was enabled, the activation request was submitted for administrative approval.

The request was reviewed and approved by the administrator. John Smith was subsequently able to activate the Security Administrator role for the configured activation period.



9.5 PIM Audit History

PIM audit history was reviewed to verify the privileged access workflow.

The audit history recorded the relevant administrative events, including the eligible role assignment, approval of the activation request, and activation of the Security Administrator role.

The audit records demonstrated that privileged operations were traceable to specific users and actions, providing an auditable record of privileged-access activity.

My audit history | Microsoft Entra roles

Privileged Identity Management | My audit history

Activity

Search by member name

Time span: Last week | Audit type: All | Subject type: All

Time	Requestor	Action	Resource name	Scope	Primary target	Subject
27/8/2026, 4:49:46 pm	Azure AD PIM	Remove member from role (PIM acti	Microserve	Microserve	Security Administrator	John Smith
27/8/2026, 2:49:47 pm	John Smith	Add member to role request approved (PIM	Microserve	Microserve	Security Administrator	John Smith
27/8/2026, 2:49:39 pm	Satya Rellu	Add member to role request approved	Microserve	Microserve	Security Administrator	John Smith
27/8/2026, 2:43:13 pm	Satya Rellu	Update role setting in PIM	Microserve	-	Security Administrator	-
27/8/2026, 2:42:05 pm	Satya Rellu	Add eligible member to role in PIM c	Microserve	Microserve	Security Administrator	John Smith
27/8/2026, 2:42:05 pm	Satya Rellu	Remove eligible member from role in	Microserve	Microserve	Security Administrator	John Smith
27/8/2026, 2:42:04 pm	Satya Rellu	Remove member from role in PIM cc	Microserve	Microserve	Security Administrator	John Smith
27/8/2026, 2:42:04 pm	Satya Rellu	Process role update request	Microserve	Microserve	Security Administrator	John Smith
27/8/2026, 1:36:19 pm	Satya Rellu	Add eligible member to role in PIM c	Microserve	Microserve	Security Administrator	John Smith
27/8/2026, 1:36:19 pm	Satya Rellu	Add eligible member to role in PIM r	Microserve	Microserve	Security Administrator	John Smith
27/8/2026, 1:34:47 pm	Satya Rellu	Update role setting in PIM	Microserve	-	Security Administrator	-

9.6 PIM Security Model

The PIM implementation demonstrated a just-in-time privileged access model:

Eligible → Request Activation → Approval → Activation → Temporary Privileged Access → Audit

This approach minimizes permanent administrative privileges and provides additional controls around privileged operations.

10. Device Identity & Endpoint Access

10.1 Device Identity in Microsoft Entra ID

Microsoft Entra device identity settings were reviewed and configured to demonstrate how organizational devices can be registered and managed within the identity platform.

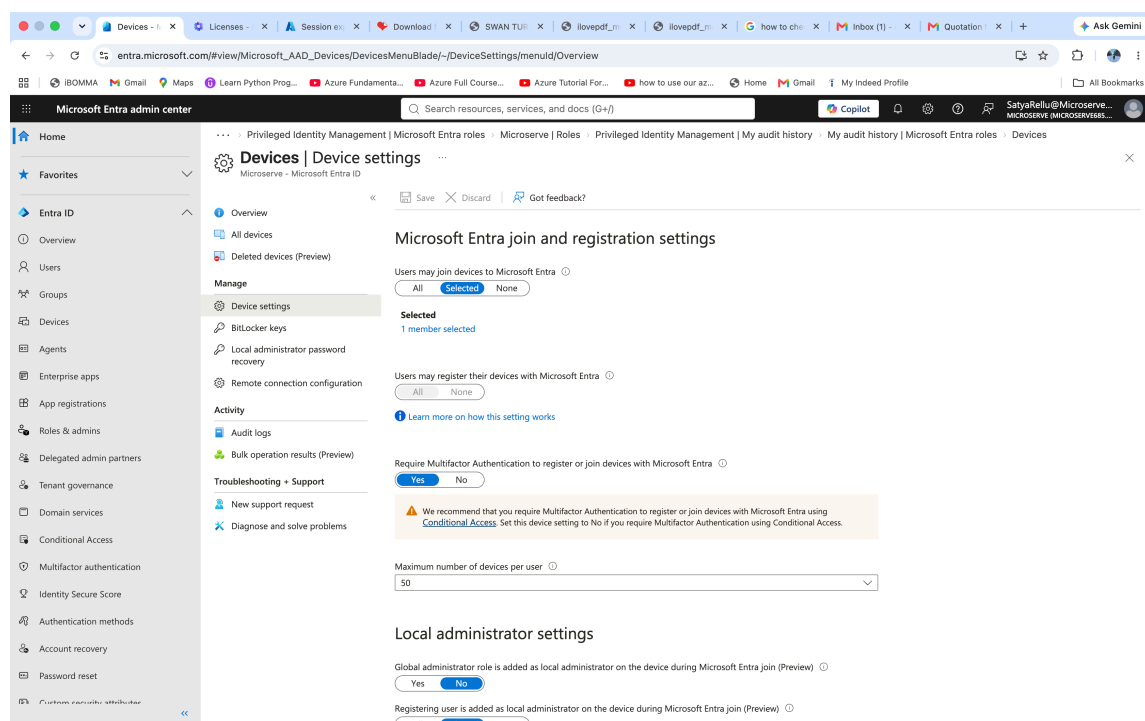
Device identity allows Microsoft Entra ID to recognize devices as part of the access-control decision. Device information can subsequently be used by Conditional Access to apply controls based on the security state of the device.

10.2 Device Join and Registration Settings

The Microsoft Entra device settings were configured to control which users can join or register devices with the organization.

The lab configuration allowed selected users to join devices to Microsoft Entra ID, with **John Smith** selected for device-join permissions. Device registration was enabled for users in the tenant.

Multifactor authentication was not directly required through the device settings because MFA requirements were handled through Conditional Access policies.



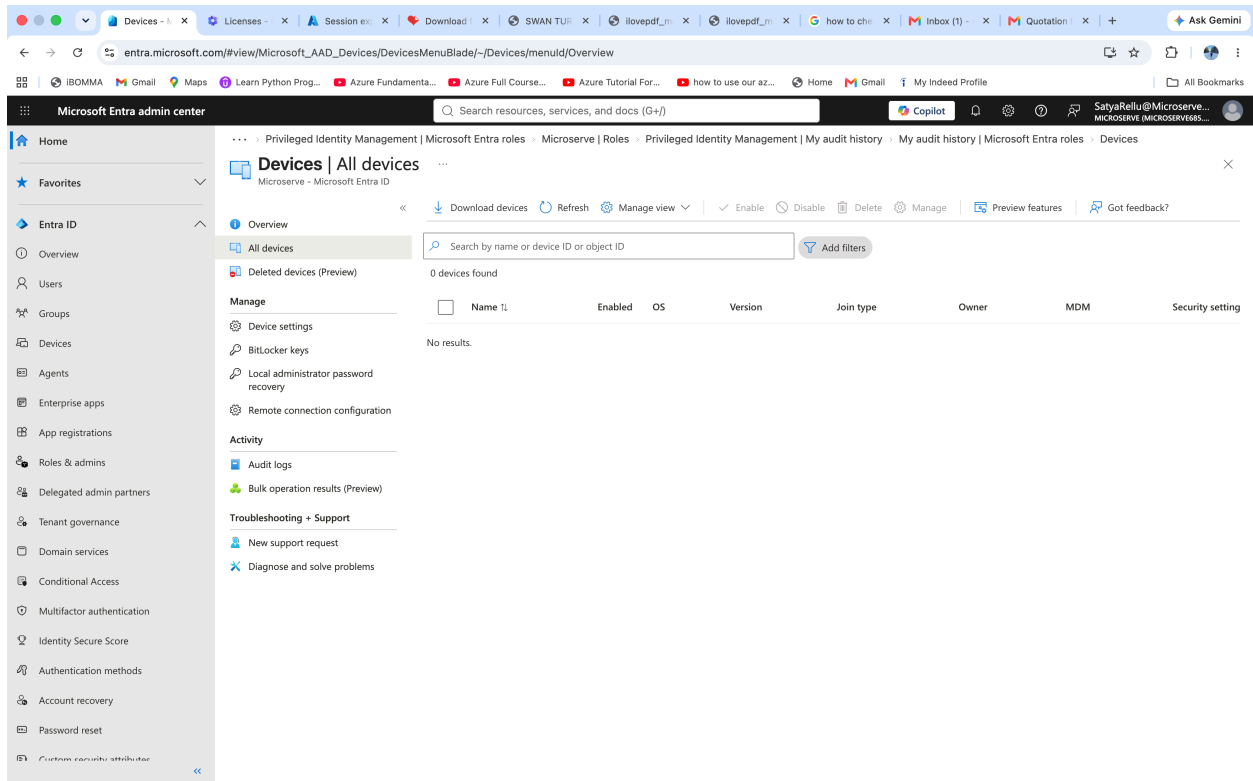
10.3 Device Limits and Local Administrator Controls

Additional device-management settings were reviewed to demonstrate control over device ownership and local administrative privileges.

The maximum number of devices per user was configured as **50**. Automatic addition of Global Administrators as local administrators during device joining was disabled.

The setting that would make the registering user a local administrator was also restricted by leaving the corresponding group assignment empty.

These controls help limit unnecessary administrative privileges on organizational devices.



10.4 Device Compliance and Conditional Access

Device compliance was incorporated into the Conditional Access design through the policy **CA010 - Require Compliant Device for Expense Portal**.

The policy was configured to require a device to be marked as compliant before access to the Microserve Expense Portal is granted. The policy was maintained in **Report-only** mode during the lab.

Because the lab environment did not contain an Intune-managed compliant device, an end-to-end compliant-device access test could not be performed.

10.5 Device Validation

The Microsoft Entra tenant was reviewed during validation and contained **no registered devices** at the time of testing.

Consequently, device registration, device compliance, and device-based Conditional Access could be configured and reviewed but could not be demonstrated through a complete device-based access scenario.

This limitation was documented as part of the lab rather than treating the absence of a managed device as a configuration failure.

11. Project Results & Conclusion

11.1 Project Outcomes

The Microsoft Entra ID IAM and Identity Governance lab successfully demonstrated the implementation of a structured identity and access management solution for the fictional Microserve organization.

The project covered the complete identity lifecycle, from user and group management through application access, authentication, Conditional Access, identity-risk monitoring, entitlement management, privileged access, and device identity.

The lab also demonstrated practical access-governance workflows, including access package requests and approvals, access assignments, access reviews, and just-in-time privileged role activation through Microsoft Entra PIM.

11.2 Security Controls Implemented

The following security and identity controls were implemented or evaluated during the project:

- Microsoft Entra user and group management
- Role-based access control
- Application registration and application roles
- Enterprise application access management
- Conditional Access
- Multifactor authentication
- Self-Service Password Reset
- Microsoft Entra Identity Protection
- Entitlement Management
- Access Packages
- Access Requests and Approvals
- Access Reviews
- Privileged Identity Management
- Just-in-time privileged access
- Device identity and device-join controls
- Sign-in and audit log monitoring

11.3 Validation Summary

The project included practical validation of the configured identity and security controls.

The Expense Portal access-governance workflow was successfully demonstrated through an access request submitted by Alex Martin, administrative approval of the request, and the resulting access package assignment.

Privileged access was successfully demonstrated through John Smith's eligible Security Administrator assignment, activation request, approval, role activation, and PIM audit history.

Sign-in and audit logs were reviewed to provide visibility into authentication and administrative activity.

Identity Protection was also reviewed, although no risky users or risky sign-ins were generated during the lab.

Conditional Access policies were configured and evaluated. The device-compliance scenario was maintained in Report-only mode because the lab did not contain an Intune-managed compliant device.

11.4 Lab Limitations

The lab environment was intentionally designed as a lightweight Microsoft Entra ID environment and did not include a Windows Server Active Directory infrastructure or an Intune-managed Windows device.

As a result, certain scenarios could be configured and reviewed but could not be demonstrated through complete end-to-end testing.

In particular, device compliance and device-based Conditional Access could not be fully validated because no managed compliant device was available in the tenant.

Similarly, the Identity Protection environment did not generate risky user or risky sign-in events during testing.

These limitations were documented as part of the project to distinguish between controls that were successfully implemented and controls that were configured but could not be fully exercised within the available lab infrastructure.

11.5 Skills Demonstrated

This project demonstrates practical knowledge of Microsoft Entra ID and identity security concepts relevant to the Microsoft SC-300 Identity and Access Administrator role.

Key skills demonstrated include:

- Identity and user management
- Group-based access management
- Role-based access control
- Application registration and enterprise applications
- Application roles and authorization
- Conditional Access policy design
- MFA and authentication controls
- Identity Protection and risk investigation
- Identity Governance
- Entitlement Management
- Access Packages and approval workflows
- Access Reviews
- Privileged Identity Management
- Just-in-time privileged access
- Audit and sign-in log analysis
- Device identity configuration
- Security-focused troubleshooting and validation

11.6 Conclusion

This project provided a practical implementation of Microsoft Entra ID identity and access management capabilities within a simulated enterprise environment.

Rather than focusing only on individual Microsoft Entra features, the project demonstrated how identity, authentication, authorization, governance, privileged access, and monitoring can work together as an integrated security model.

The Microserve Expense Portal scenario provided a consistent business context for applying these controls and demonstrating realistic access-management workflows.

The project also identified the boundaries of the lab environment and documented areas where additional infrastructure, such as Microsoft Intune or hybrid Active Directory, would be required for deeper end-to-end testing.